Assignment 5

Module -5: Network Fundamentals and Building Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?

a) Assigning IP addresses to devices

b) Providing wireless connectivity to devices

c) Forwarding data packets between networks

d) Managing user authentication and access control

Ans : c) Forwarding data packets between networks

2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?

a) Assigning static IP addresses to devices

b) Resolving domain names to IP addresses

c) Managing network traffic and congestion

d) Dynamically assigning IP addresses to devices

Ans : d) Dynamically assigning IP addresses to devices

3. Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?

a) Router

b) Switch

c) Hub

d) Repeater

Ans : b) Switch

4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?

a) Star

b) Bus

c) Ring

d) Mesh

Ans : b) Bus

Section 2: True or False

5. A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

Ans : True

6. TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

Ans : False

7. A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Ans : True

Section 3: Short Answer

8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Ans :

1. Connect modem to router
2. Power on router
3. Connect computer to router
4. Open browser and enter router IP
5. Login using admin credentials
6. Set WiFi name (patel) and password (7776)
7. Enable security (WPA2/WPA3)
8. Enable DHCP
9. Save and restart

- After you do all this steps your network is ….

  Secure

  Accessible

  Automatically configured (via DHCP)

Section 4: Practical

9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

Ans :

1. Connect router to network and access CLI
2. Enter in configuration mode (enable, configure terminal)
3. Configure LAN interface with IP address
4. Exclude reserved IP addresses
5. Create DHCP pool and define network
6. Set default gateway and DNS server
7. Configure interface to get IP using DHCP
8. Save the configuration

❖ Example

```
en
conf t

interface FastEthernet0/0
ip address 10.0.0.1 255.0.0.0
no shutdown
exit
do wr

ip dhcp pool patel
network 10.0.0.0 255.0.0.0
default-router 10.0.0.1
exit
do wr

ip dhcp excluded-address 10.0.0.1 10.0.0.10
do wr
```

✓ after that your pc get a automatically ip.

Section 5: Essay

10. Discuss the importance of network documentation in the context of building and managing networks.

Ans :

- Network documentation means keeping records of:

  1. Network design
  2. Devices
  3. IP addresses
  4. Configurations

  ❖ Importance:

  1. **Troubleshooting :**
     Helps quickly identify and fix problems
  2. **Network Management :**
     Makes it easy to understand the system
  3. **Security :**
     Tracks devices and access control
  4. **Time Saving :**
     Faster maintenance and upgrades
  5. **Disaster Recovery :**
     Helps restore network after failure
  6. **Scalability :**
     Easy to expand network in future

✓ Without documentation, managing a network becomes difficult.
✓ It ensures smooth, secure, and efficient operation.